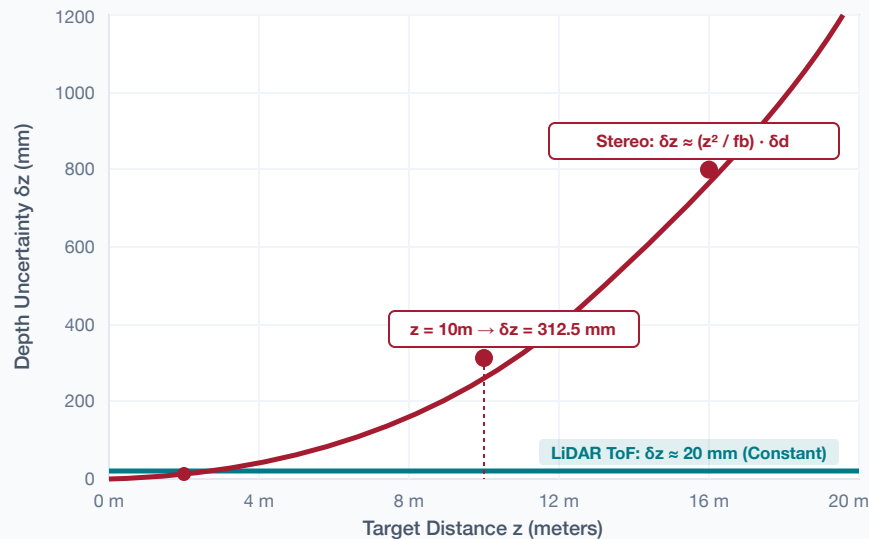


METRIC GROUNDING, DEPTH ERROR SCALING & CLEARANCE INSETS

Quadratic Stereo Divergence vs Constant LiDAR ToF Error and Its Direct Subtraction from Physical Stopping Clearance

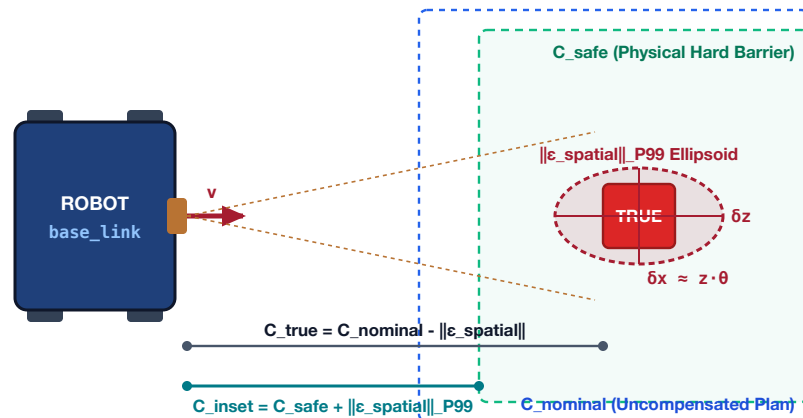
A. DEPTH UNCERTAINTY DIVERGENCE (δz vs RANGE z)



Nonlinear Unprojection & Inverse Depth Formulation:

- Standard stereo mapping $z = f/d$ maps Gaussian disparity noise into a heavy-tailed distribution in 3D
- Doubling range quadruples depth uncertainty (**quadratic divergence**).
- State estimators use **inverse depth** $p = 1/z$ coordinates to preserve linear Kalman updates as $d \rightarrow 0$.

B. PROPAGATED ERROR & CLEARANCE INSET GEOMETRY



BEDROCK INVARIANT: UNINSET CLEARANCE MARGINS ARE A FICTION

The Mathematical Inset Rule: To guarantee physical separation C_{safe} , the safety enforcer must enforce $C_{\text{inset}} = C_{\text{safe}} + \|\epsilon_{\text{spatial}}\|_{\text{P99}}$ where $\epsilon_{\text{spatial}}$ includes sensor depth noise and extrinsic

Collision Inevitability without Insetting:

If nominal margin is 100mm but perception carries 150mm uncertainty (P99), the physical mass will strike the obstacle while the software monitor asserts nominal clearance is safe.